GTU Internship 2026 Problem Statements

Problem Domain 1: Border Defence and Surveillance

Border security is a critical national priority, requiring constant monitoring and rapid response to potential threats. Traditional surveillance systems often rely on manual monitoring, which can be inefficient and error-prone.

Key challenges include:

- **Large-Scale Monitoring**: Vast border areas generate enormous amounts of surveillance data.
- **Delayed Threat Detection**: Manual analysis can result in late identification of intrusions or suspicious activity.
- **False Alarms**: Animals, weather conditions, or environmental noise can trigger unnecessary alerts.
- **Resource Constraints**: Limited manpower and equipment must cover extensive and remote regions.
- **Data Integration Issues**: Sensor, satellite, and historical data are often siloed.

Objectives

Students can:

- Perform **EDA on surveillance and sensor datasets**
- Build **anomaly detection models** to identify unusual activities
- Classify objects or movement patterns using **ML/DL techniques**
- Analyze historical incident data to predict high-risk zones
- Develop alert prioritization systems to reduce false positives

Dataset Availability
Open datasets related to **satellite imagery, sensor data, object detection, and simulated surveillance data** are publicly available.